

Sectorization of Water Distribution Systems for Optimal Observability[★]

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Abstract: Resilient water distribution systems have the ability to anticipate, monitor, react to and learn from critical events. While numerous resilience metrics have been proposed for the reacting and anticipating functions, monitoring metrics are understudied. This work proposes an a-priori monitoring metric based on structural observability theory, the minimum number of sensors, and demonstrates how it can be integrated into problem formulations for optimal sectorization. The dividing stage of WDS sectorization is modelled as a MINLP, while the minimum sensor placement problem is solved using zero forcing formulated as an IP. To provide an example, the methodology is applied to the benchmark network Apulia.

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1. INTRODUCTION

Water distribution systems (WDSs) are sensitive to critical events due to aging, progressing urbanization and to extreme weather events expected to become more frequent due to climate change. As critical infrastructure securing access to safe water in many cities, they should be able to withstand and recover from critical events, i.e. be resilient. Resilient WDSs have the ability to anticipate, monitor, react and learn (Leštáková et al., 2024). While numerous resilience metrics have been proposed for the reacting and anticipating functions, monitoring metrics are understudied (Leštáková et al., 2024). Such metrics would be beneficial for WDS sectorization studies which often argue that sectorized WDSs are easier to monitor (Vasilic et al., 2024; Di Nardo et al., 2018; Perelman and Ostfeld, 2011).

Existing monitoring metrics rely on a-posteriori analysis, quantifying, for example, the detection time of a critical event such as leakage (Mounce et al., 2012; Romano et al., 2014), minimum detectable failure quantity value such as leakage outflow (Du et al., 2024), or the ability to localize a critical event in terms of distance (Romero-Ben et al., 2022) for a specific set of sensors. A metric able to quantify how easy it is to monitor a system a-priori would allow embedding it in network design and analysis algorithms, helping practitioners adjust existing networks accordingly and design new networks so that they are inherently easy to monitor.

We aim to develop such a-priori monitoring metric based on structural observability theory. Observability is a system property that informs about the possibility of determining a system's state from its outputs. In the domain of WDSs, observability has been used e.g. for state estimation (Díaz et al., 2016) or for water quality sensor placement (Kazma et al., 2024). However, these approaches lack scalability and require exact knowledge of the model parameters that are often subject to uncertainties (van Gemert et al., 2024). van Gemert et al. (2024) have shown that these issues can be circumvented by exploiting structural observability theory, applying an approach developed by Jia et al. (2021) to WDS.

Observability is based on a crisp definition: the system either is or is not observable, making it poorly suited to serve as a metric (Montanari and Aguirre, 2020). Some a-posteriori metrics have been proposed to quantify how well a system is observable using a specific set of outputs (e.g. by Friedland (1975)); however, this is practical only if the system is observable in the first place and cannot be used to assess how “unobservable” a system is. This motivates the use of minimal number of sensors that would render a WDS observable as a proxy to describing how well a system can be monitored in the scope of this work. The evaluation of the monitoring metric thus becomes a minimum sensor placement problem.

Sensor placement problems in WDS have received considerable scientific attention. They mostly focus on determining the optimal sensor positions for monitoring water quality (e.g. de Winter et al. (2019); Kazma et al. (2024) with limited number of sensors) or leakage detection. Examples for the latter include Du et al. (2024) who proposed a multi-objective pressure sensor placement method for maximizing burst detection while minimizing the number of sensors, and Romero-Ben et al. (2022) performing model-free optimal sensor placement of a fixed number of

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sensors for best possible leakage localization. Both of these papers use evolutionary algorithms to solve the respective optimization problem. Sela and Amin (2018) provide a MIP formulation to the sensor placement problem for leakage detection using a limited number of sensors with predefined fault detection matrix, maximizing detectability. In contrast to these approaches, the presented work uses the minimal number of sensors as a monitoring metric for WDS rather than striving to determine the optimal sensor location for a fixed or variable number of sensors. We demonstrate how the computation of this metric can be formulated as an integer program and integrated into an optimal sectorization problem formulated as a MINLP, avoiding the use of evolutionary algorithms.

The main contributions of this work are the following:

- providing a MINLP formulation for the dividing stage of WDS sectorization
- proposing an observability metric based on structural observability theory and showing how it can be solved to optimality using zero forcing
- demonstrating how the observability metric can be integrated in the sectorization MINLP to deliver an optimally observable sectorization

2. PRELIMINARIES

2.1 Structural Observability for WDSs

Observability determines whether it is possible to reconstruct the trajectory temporal evolution of the internal states of a system from the knowledge of inputs and a given set of outputs (Montanari and Aguirre, 2020; van Gemert et al., 2024). Designed with low-dimensional systems in mind, classic control theory methods for assessing observability like the Kálmán rank test or the Hautus test are not feasible for large-scale systems such as WDSs (Montanari and Aguirre, 2020). To address this issue, the concept of structural observability has been proposed, working with the assumption that in a network modelled as a mathematical graph, the topology of the graph, i.e. the presence of edges between nodes, is known, but the edge weights are unknown (Jia et al., 2021). This corresponds well to a common problem in modelling of WDSs: the topology of the WDS is generally known, but the parameters defining the edges such as pipe roughness or even pipe lengths and diameters are often subject to uncertainty (van Gemert et al., 2024).

This work relies on the theoretical foundations laid out by Jia et al. (2021) for structural controllability that have been reformulated for the dual problem of structural observability on the example of water distribution networks by van Gemert et al. (2024). Below, the methodology behind assessing structural observability of WDS developed by van Gemert et al. (2024) is briefly summarized. An interested reader can consult van Gemert et al. (2024) as well as Jia et al. (2021) and Shali et al. (2022) for detailed description and proofs.

A continuous-time, autonomous, time-invariant, nonlinear system

$$\begin{aligned} \dot{x}(t) &= f(x(t)), & t \in \mathbb{R}^+, \\ y(t) &= Cx(t), \end{aligned} \quad (1)$$

where $x(t) \in \mathbb{X} \subseteq \mathbb{R}^{n_x}$ is the state vector, $y(t) \in \mathbb{R}^{n_y}$ the output vector, f a nonlinear function matrix and $C \in \mathbb{R}^{n_y \times n_x}$ the output measurement matrix, can be linearized around an arbitrary operation point x_0 as follows:

$$\begin{aligned} \delta \dot{x}(t) &= A(x_0)\delta x(t), & t \in \mathbb{R}^+, \\ \delta y(t) &= C\delta x(t), \end{aligned} \quad (2)$$

with $\delta x(t) = x(t) - x_0$ and $A(x_0) \in \mathbb{R}^{n_x \times n_x}$ being the system matrix (van Gemert et al., 2024). According to the classic definition of observability by Kálmán, a system is said to be observable if, for any unknown initial state $x_0 \in \mathbb{X}$, there exists a finite time such that the output y suffices to uniquely determine $x(0)$ (Kalman, 1960). Van Gemert et al. extend this definition to nonlinear systems, according to which a nonlinear system (Eq. 1) is called locally observable around x_0 if the linearized system (Eq. 2) derived from it at x_0 is observable, and, moreover, the nonlinear system is called observable if it is locally observable for all $x_0 \in \mathbb{X}$ (van Gemert et al., 2024, Def. II.2).

In systems like WDSs, the precise values of entries in the system matrix A are often not known, but the underlying structure is known exactly. Such systems can be studied using pattern matrices which only provide information about whether a specific entry is a fixed zero (0), non-zero (*), or arbitrary (?) value. To formalize this, consider the pattern class $\mathcal{P}$ of the $p \times q$ pattern matrix $\mathcal{M}$, defined as:

$$\mathcal{P}(\mathcal{M}) := \{M \in \mathbb{R}^{p \times q} \mid \begin{aligned} M_{ij} &= 0 & \text{if } \mathcal{M}_{ij} = 0, \\ M_{ij} &\neq 0 & \text{if } \mathcal{M}_{ij} = *, \\ M_{ij} &\in \mathbb{R} & \text{if } \mathcal{M}_{ij} = ? \end{aligned} \}. \quad (3)$$

For system matrix A and the output matrix C from (2) one can define pattern matrices $\mathcal{A} \in \{0, *, ?\}^{n_x \times n_x}$ and $\mathcal{C} \in \{0, *, ?\}^{n_y \times n_x}$ such that $A \in \mathcal{P}(\mathcal{A})$ and $C \in \mathcal{P}(\mathcal{C})$. The corresponding structured system will be denoted $(\mathcal{A}, \mathcal{C})$. The structured system $(\mathcal{A}, \mathcal{C})$ is called strongly structurally observable if the pair (A, C) is observable for all $A \in \mathcal{P}(\mathcal{A})$ and $C \in \mathcal{P}(\mathcal{C})$ (van Gemert et al., 2024, Def. II.4). Assessment of the observability of the structured system then follows from the Hautus test (Shali et al., 2022), aiming to determine whether the pattern matrices $\mathcal{M} = [\mathcal{A}^\top \mathcal{C}^\top]$ and $\bar{\mathcal{M}} = [\bar{\mathcal{A}}^\top \bar{\mathcal{C}}^\top]$ have full row rank. Here,

$$\bar{\mathcal{A}}(i, i) := \begin{cases} * & \text{if } \mathcal{A}(i, i) = 0 \\ ? & \text{otherwise.} \end{cases} \quad (4)$$

The rank determination is performed by first using the above matrices $\mathcal{M}$ and $\bar{\mathcal{M}}$ to construct corresponding associated graphs $\mathcal{G}^z = \{\mathcal{E}^z, \mathcal{V}^z\}$ and $\bar{\mathcal{G}}^z = \{\bar{\mathcal{E}}^z, \bar{\mathcal{V}}^z\}$ which capture the interactions (represented by graph edges) between all state variables (represented by graph nodes). An edge $(j, i) \in \mathcal{E}^z$ if $\mathcal{M}(i, j) = *$ (same holds for $\bar{\mathcal{M}}$ and $\bar{\mathcal{G}}^z$). Subsequently the colorability of the associated graph is assessed. Colorability assessment is a method to determine the minimum rank of graphs, and several color change rules have been proposed based on the characteristics of the studied graph (Hogben, 2010). The color change rule adopted for WDSs by van Gemert et al. (2024) from Jia et al. (2021) can be summarized as follows: if exactly one out-neighbor v of the node u is white, then change the color of v to blue (“ u forces v ”). If, after applying the rule until no more changes are possible, all nodes corresponding to $\mathcal{A}$ have been colored blue, the

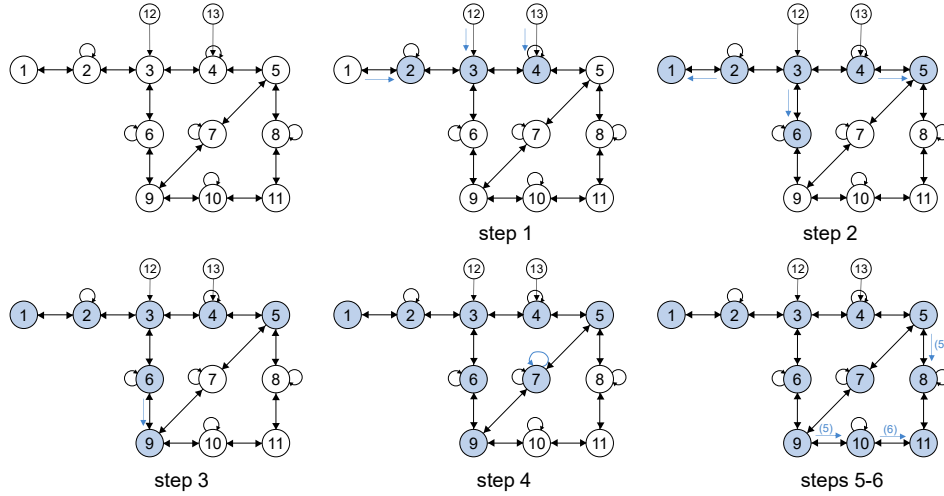


Fig. 1. Applying the color change rule introduced in Section 2.1 to the graph representation of the matrix in Eq. 5. The nodes 12 and 13 result from the columns of matrix $\mathcal{M}$ corresponding to the output matrix $\mathcal{C}$ and will always have precisely one out-neighbor. Together with node 1, they can force nodes 3, 4 and 2, respectively, in step 1. Remaining steps demonstrate the further propagation of the color change rule, showing that the graph is colorable as all nodes corresponding to $\mathcal{A}$ are blue. Nodes 12 and 13 constitute the zero forcing set of the graph.

graph is considered colorable, and the matrix $\mathcal{M}$ is said to have full rank (Jia et al., 2021, Th. 10). This way, it can be shown whether output states specified via the output matrix $\mathcal{C}$ correspond to a zero forcing set of the associated graph (a set of nodes that, if initially colored, will render the graph colorable). Finally, if the graphs $\mathcal{G}^z$ and $\bar{\mathcal{G}}^z$ are colorable, the structured system $(\mathcal{A}, \mathcal{C})$ is strongly structurally observable (van Gemert et al., 2024, Def. IV.3).

An example of applying the color change rule to the graph representation of the matrix from Eq. 5 is given in Fig. 1. In Eq. 5, the dashed line in matrix $\mathcal{M} = [\mathcal{A}^\top \mathcal{C}^\top]$ separates the parts corresponding to $\mathcal{A}^\top$ and $\mathcal{C}^\top$.

$$\mathcal{M} = \begin{matrix} & \begin{matrix} 2 & 4 & 6 & 7 & 8 & 10 & 1 & 3 & 5 & 9 & 11 \end{matrix} & \begin{matrix} 12 & 13 \end{matrix} \\ \begin{matrix} 2 \\ 4 \\ 6 \\ 7 \\ 8 \\ 10 \\ 1 \\ 3 \\ 5 \\ 9 \\ 11 \end{matrix} & \begin{bmatrix} * & 0 & 0 & 0 & 0 & 0 & * & * & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & * & 0 & 0 & 0 & 0 & 0 & * & * & 0 & 0 & 0 & 0 & * \\ 0 & 0 & * & 0 & 0 & 0 & 0 & * & 0 & * & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & * & 0 & 0 & 0 & 0 & * & * & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & * & 0 & 0 & 0 & * & 0 & * & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & * & 0 & 0 & 0 & * & * & 0 & 0 & 0 \\ * & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ * & * & * & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & * & 0 \\ 0 & * & 0 & * & * & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & * & * & 0 & * & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & * & * & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \end{matrix} \quad (5)$$

van Gemert et al. (2024) show how observability analysis by colorability assessment can be applied to WDSs modelled with the elastic water column model proposed by Zeng et al. (2022). This transient WDS model combines the advantages of rigid water column models used for simulation of slow transients and an electrical equivalent model that allows capturing water compressibility, avoiding the need of spatial discretization (Zeng et al., 2022). Crucially, the governing equations are easy to formulate in state-space form. Upon linearization around an operation point q_0 and h_0 , we obtain

$$\begin{bmatrix} \frac{dq}{dt} \\ \frac{dh}{dt} \end{bmatrix} = \begin{bmatrix} -L^{-1}R \text{diag} \left\{ \frac{2q_0^2}{|q_0|} \right\} & L^{-1}A_{\text{inc}}^\top \\ F^{-1}A_{\text{inc}} & -F^{-1}D \text{diag} \left\{ \frac{1}{2\sqrt{h_0}} \right\} \end{bmatrix} \begin{bmatrix} q \\ h \end{bmatrix} \quad (6)$$

with the state variable vectors q and h of size m and n representing volume flow and hydraulic head, respectively. In Eq. 6, $L = \text{diag}\{l/(gS)\}$ is the hydraulic inductance matrix, $R = \text{diag}\{fl/(2gDS^2)\}$ the hydraulic resistance matrix, $F = \text{diag}\{1/2A_{\text{inc}}C + CT\}$ with the hydraulic capacitance matrix $C = \text{diag}\{gAl/a^2\}$ and tank capacitance $CT = \text{diag}\{S\}$, and last, the discharge matrix $D = \text{diag}\{C_d S_d \sqrt{2}q\}$.

The system matrix in Eq. 6, denoted $A(x_0)$, can be represented by the pattern matrix

$$\mathcal{A}_{\text{WDS}} = \begin{bmatrix} \mathcal{I} & \mathcal{A}_{\text{inc}}^\top \\ \mathcal{A}_{\text{inc}} & \mathcal{D} \end{bmatrix} \quad (7)$$

such that $A(x_0) \in \mathcal{P}(\mathcal{A}_{\text{WDS}})$ (van Gemert et al., 2024). Above, $\mathcal{I} = \text{diag}\{*\} \in \{0, *\}^{m \times m}$ (pattern identity matrix), $\mathcal{D} = \text{diag}\{?\} \in \{0, ?\}^{n \times n}$ (pattern discharge matrix), and $\mathcal{A}_{\text{inc}} \in \{0, *\}^{n \times m}$ is a pattern representation of the incidence matrix A_{inc} of the WDS graph. Performing structural observability analysis using the pattern matrix $\mathcal{A}_{\text{WDS}}$ means that instead of the knowledge of the parameters in the system matrix $A(x_0)$ in Eq. 6, only the underlying WDS graph topology given via its incidence matrix A_{inc} is required.

2.2 Zero Forcing for Structural Observability

Hogben (2010) states that a subset of vertices that, if originally colored blue, will render a graph colorable, is the zero forcing set of a graph. The zero forcing number is the minimum size of all zero forcing sets of the graph. Finding the minimum number of sensors, i.e. the optimal $\mathcal{C}$ for rendering the structured system $(\mathcal{A}, \mathcal{C})$ observable, is then equivalent to finding the zero forcing number and the corresponding zero forcing set of the associated graphs of $\mathcal{A}$ and $\bar{\mathcal{A}}$ (van Gemert et al., 2024, Def. IV.3). This problem

can be solved algorithmically like in van Gemert et al. (2024). We show that it can also be solved by an integer program formulated by Brimkov et al. (2018), which allows it to be integrated into MI(N)LPs more efficiently.

Let $\mathcal{G}^z = \{\mathcal{V}^z, \mathcal{E}^z\}$ denote a graph with the set of vertices $\mathcal{V}^z$ and edges $\mathcal{E}^z$.

$$\min \sum_{v \in \mathcal{V}^z} s_v \quad (8)$$

$$\text{s.t. } s_v + \sum_{e \in \delta^-(v)} y_e \leq 1 \quad \forall v \in \mathcal{V}^z \quad (9)$$

$$x_w - x_v + (\tau + 1)y_e \leq \tau \quad \forall e = (u, v) \in \mathcal{E}^z, \quad \forall w \in N(u) \setminus \{v\} \quad (10)$$

$$x_u - x_v + (\tau + 1)y_e \leq \tau \quad \forall e = (u, v) \in \mathcal{E}^z \quad (11)$$

The binary variable s_v indicates whether the vertex v is in the zero forcing set, i.e. whether a sensor is to be placed on it. The size of the zero forcing set, i.e. the zero forcing number $Z = \sum_{v \in \mathcal{V}^z} s_v$ is to be minimized. y_e is a binary variable for each edge $e = (u, v)$ indicating whether u forces v . x_v are integer variables ranging in $\{0, \dots, \tau\}$ that indicate at which timestep the vertex v is forced (colored). τ is the maximum difference between the forcing times of two vertices, also referred to as propagation time. $\delta^-(v)$ denotes the set of edges point towards node v , and $N(u)$ is the set of out-neighbor vertices of u . (Brimkov et al., 2018)

The first constraint in Eq. 9 requires every vertex v to be either a member of the zero forcing set ($s_v=1$) or be forced by precisely one of the edges leading to it.

The constraint in Eq. 10 prescribes the order in which the vertices are forced. If the vertex u is to force v (i.e. $y_e=1$), each out-neighboring vertex w of u other than v would already have to be forced before v . In other words, a vertex v can be forced only if it is the only white out-neighbor of u .

The constraint in Eq. 11 expresses that if the vertex u is to force v (i.e. $y_e=1$), the vertex u would have to be forced before v ($x_u \leq x_v - 1$). In other words, only a blue vertex can force.

Upon solving, the IP will deliver the zero forcing number Z required to make the graph corresponding colorable, i.e. to make the system characterized by the system matrix A observable.

2.3 WDS Modelling and Sectorization

Dividing WDSs into smaller parts (sectors) belongs to the most attractive strategies for WDS management and has received considerable attention in research (Di Nardo et al., 2018). It is performed most commonly for the purpose of leakage detection and management, pressure monitoring and management, monitoring of water quality (Di Nardo et al., 2018).

The problem of dividing a WDS into smaller sectors (sectorization) most commonly entails determining the sizes and shapes of the sectors and altering the network topology by closing suitable edges connecting the sectors. This problem is combinatorially large and is usually solved

in two stages: (a) node clustering stage and (b) dividing stage Di Nardo et al. (2018); Vasilic et al. (2024).

Node clustering can be performed using graph clustering algorithms (e.g., community detection (Giustolisi and Ridolfi, 2014), identification of strongly connected components (Vasilic et al., 2024), spectral clustering (Di Nardo et al., 2018)), graph partitioning algorithms (such as METIS (Di Nardo et al., 2018)), or other algorithms designed specifically for this purpose (Di Nardo et al., 2015). Ideally, the algorithm allows for considering the following criteria specified by the IWA (IWA, 2007):

- sectors should be contiguous
- sectors should have similar sizes in terms of demand
- nodes in one sector should have similar elevation and/or demand
- boundary edges between sectors should not entail important edges, i.e. they should have smallest possible volume flow

Of the algorithms stated above, METIS is able to ensure sector contiguity and allows clustering with multiple weights (e.g. demand and elevation) while minimizing the number of edges between sectors. It has been shown to deliver satisfactory results for WDSs (Di Nardo et al., 2018).

The dividing stage, the aim of which is to determine which boundary edges can be closed, is most commonly performed using black box optimization methods such as evolutionary algorithms (Zeidan et al., 2021; Giustolisi and Ridolfi, 2014). The advantage of this method is that it allows for embedding conventional solvers, simulating prolonged periods of time, and easy integration of multiple objectives. While good at finding feasible solutions, however, it is computationally intensive and lacks optimality guarantees. In contrast, conventional optimization methods like MI(N)LP require more modeling effort but can deliver globally optimal solutions.

3. MAIN RESULTS

This section presents the main contribution of this work, namely the formulation of the sectorization problem as a MINLP (Sec. 3.1) and integrating the observability metric into it (Sec. 3.2) to deliver an optimization program for WDS sectorization for optimal observability, measured by the proposed monitoring metric (minimum number of sensors).

3.1 Sectorization Problem as a MINLP

In the sectorization problem, we assume that the result of the node clustering stage, i.e. which nodes belong to which sector as well as which edges lie between different sectors (boundary edges), is known.

For the dividing stage, we propose a MINLP formulation that maximizes the number of divided boundary edges while satisfying demand during a single time step under quasi-static flow constraints. The boundary edges are modeled by the binary variable b_{ij} (1 if edge (i, j) is divided):

$$\max \sum_{(i,j) \in E} b_{ij}. \quad (12)$$

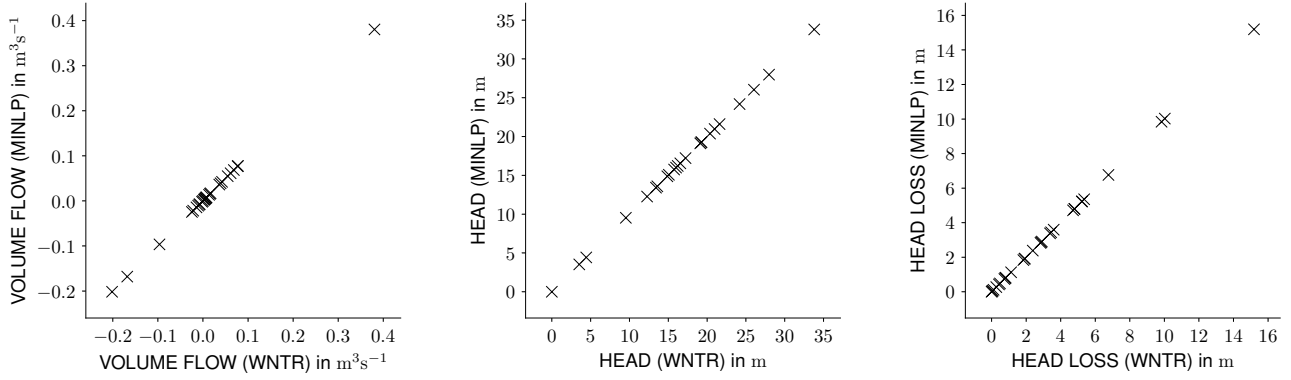


Fig. 2. Comparison between results of results obtained using WNTR (x -axis) and the proposed MINLP (y -axis) for the network Apulia divided into 2 sectors.

The program is based on the quasi-static model of WDS flow. The constraints embodying the mass conservation, energy conservation and Hazen-Williams pressure loss equations are, respectively:

$$\sum_{(i,v) \in E} q_{iv} - \sum_{(v,j) \in E} q_{vj} = Q_v \quad \forall v \in \mathcal{V} \quad (13)$$

$$\begin{aligned} p_i + \alpha_i - p_j - \alpha_j - \Delta p_{ij}(x_{ij} - y_{ij}) &\leq \Delta P^{\max} b_{ij} \\ p_i + \alpha_i - p_j - \alpha_j - \Delta p_{ij}(x_{ij} - y_{ij}) &\geq -\Delta P^{\max} b_{ij} \end{aligned} \quad (14)$$

$\forall (i, j) \in \mathcal{E}$

$$\Delta p_{ij} = 10.667 f_{ij}^{-1.852} d_{ij}^{-4.871} l_{ij} |q_{ij}|^{1.852} \quad (15)$$

The variables in constraints 13–15 (volume flow q_{ij} , pressure loss Δp_{ij} , flow direction decision variables x_{ij}, y_{ij} , edge division decision variable b_{ij} at an edge connecting node i to j , and pressure p_i at node i) as well as all necessary parameters and sets are specified in Table 1. $\Delta P^{\max}$ is a sufficiently large parameter to ensure that the energy conservation does not have to hold if the edge is divided, i.e. if $b_{ij} = 1$. $\mathcal{V}$ and $\mathcal{E}$ are the sets of nodes and edges of the WDS graph $\mathcal{G} = \{\mathcal{V}, \mathcal{E}\}$. Demand q_j has to be satisfied at nodes j belonging to the set of junctions $\mathcal{J}$:

$$q_j = Q_j \quad \forall j \in \mathcal{J}. \quad (16)$$

Only boundary edges can be divided:

$$b_{ij} \leq \beta_{ij} \quad \forall (i, j) \in \mathcal{E}, \quad (17)$$

where the boundary edge parameter $\beta_{ij} \in \{0, 1\}$ results from the node clustering phase and indicates which edges are on the boundaries between different sectors.

The volume flow q_{ij} across divided edges is zero, while it is bounded by the sufficiently large parameter $Q^{\max}$ on open edges:

$$\begin{aligned} q_{ij} &\leq (1 - b_{ij}) Q^{\max}, \\ q_{ij} &\geq -(1 - b_{ij}) Q^{\max} \end{aligned} \quad \forall (i, j) \in \mathcal{E} \quad (18)$$

Additional constraints serve to determine the flow direction (Eq. 19), to model tank in- and outflow (Eq. 20) and to enforce graph connectivity.

$$\left. \begin{aligned} x_{ij} + y_{ij} &\leq 1 \\ q_{ij} &\leq Q^{\max} x_{ij} \\ q_{ij} &\geq -Q^{\max} y_{ij} \end{aligned} \right\} \forall (i, j) \in \mathcal{E} \quad (19)$$

$$\left. \begin{aligned} 0 &\leq p_t \leq P_t^{\max} \\ p_t &= \varrho_t q_t / S_t + P_t^0 \\ q_t &\leq Q_t^{\max} \cdot \varrho_t \\ q_t &\geq -Q_t^{\max} (1 - \varrho_t) \end{aligned} \right\} \forall t \in \mathcal{T} \quad (20)$$

The solution to the network hydraulics delivered by the MINLP corresponds to that of quasi-static demand-driven simulation results for a single time step by the conventional solver for extended period steady-state simulation, WNTR (Klise et al., 2017), as shown in Fig. 2.

3.2 Including Observability in the Sectorization MINLP

The IP introduced in Sec. 2.2 can be integrated into the sectorization MINLP (Sec. 2.3) as additional constraints and an additional objective.

The IP was originally proposed for determining the zero forcing number of simple digraphs and as such it follows a different color change rule than the one proposed by Jia et al. (2021) and van Gemert et al. (2024) for distribution networks, in which the pattern matrices yield loop digraphs. The difference is that for simple digraphs, the color change rule prescribes that only a blue vertex can force another vertex, while for loop digraphs, a vertex of either color can force (Hogben, 2010). The color change rules are not interchangeable. The IP can, however, be easily adjusted too loop digraphs by omitting the constraint in Eq. 11.

In order to integrate the ZFS into the MINLP, the first step would be to create the associated graph $\mathcal{G}^w = \{\mathcal{V}^w, \mathcal{E}^w\}$ from the pattern matrix $\mathcal{A}_{\text{WDS}}$. The set of vertices of this graph would have the size of the number of state variables, i.e. volume flows at the WDS edges and pressures at WDS nodes: $|\mathcal{V}^w| = n + m$. The set of edges of the graph depends on the entries in the $\mathcal{A}_{\text{WDS}}$ pattern matrix. The dividing stage (the closing of boundary edges) would, however, alter the topology of the WDS and thus also the associated graph $\mathcal{G}^w$: the sets $\mathcal{V}^w$ and $\mathcal{E}^w$ would need to be variable.

To circumvent this modelling problem, we construct matrix variables A^w and $\bar{A}^w$ corresponding to the pattern system matrices $\mathcal{A}_{\text{WDS}}$ and $\bar{\mathcal{A}}_{\text{WDS}}$ using matrix variables $A^{\text{inc}}, I, \bar{I}$ and $D, \bar{D}$ according to Eq. 7 by using the open boundary edge matrix B^0 , constructed from the indicator of edge division b_{ij} , see Eq. 23. Dividing a WDS edge is then equivalent to setting the corresponding values of the system matrix variable to 0. This allows to keep the sets

of corresponding nodes constant during the optimization program.

Eq. 21 states the objective function: minimizing the number of sensors to place stored in the binary variable s_v^{clean} . The sum term in the objective function is the monitoring metric. Constraints in Eqs. 22 - 25 serve to construct the variable system matrices A^w and $\bar{A}^w$, and constraints in Eqs. 26-27 specify that only edges present in the system matrix variables can force.

$$\min \sum_{v \in \mathcal{V}^w} s_v^{\text{clean}} \quad (21)$$

$$\text{s.t. : } B^o = \text{diag}\{1 - b_{ij} \mid \forall (i, j) \in \mathcal{E}\} \quad (22)$$

$$A^{\text{inc}} = \alpha^{\text{inc}} B^o \quad (23)$$

$$A^w = \begin{bmatrix} I & A^{\text{incT}} \\ A^{\text{inc}} & D \end{bmatrix} \quad (24)$$

$$\bar{A}^w = \begin{bmatrix} \bar{I} & A^{\text{incT}} \\ A^{\text{inc}} & \bar{D} \end{bmatrix} \quad (25)$$

$$A_{ij}^w - y_{ij}^z \geq 0 \quad \forall i \in \mathcal{V}^w, \forall j \in \mathcal{V}^w \quad (26)$$

$$\bar{A}_{ij}^w - \bar{y}_{ij}^z \geq 0 \quad \forall i \in \mathcal{V}^w, \forall j \in \mathcal{V}^w \quad (27)$$

Constraints in Eqs. 28 and 29 are implementations of Eqs. 9 and 10, respectively. To understand the constraints in Eq. 30, recall that dividing a WDS edge is equivalent to setting the corresponding values of the system matrix variable to 0. This does not remove nodes from the associated graph: instead, they become disconnected from the rest of the graph. As such, they cannot be forced by any edge and will always belong to the zero forcing set. Eq. 30 removes these nodes from the sensor placement variable s , delivering s^{clean} .

$$s_v + \sum_{i \in \mathcal{V}^w} y_{iv}^z \leq 1 \quad \forall v \in \mathcal{V}^w \quad (28)$$

$$\left. \begin{aligned} x_k^z - x_j^z + (\tau + 1)y_{ij}^z A_{ki}^w &\leq \tau & \forall i \in \mathcal{V}^w, \\ \bar{x}_k^z - \bar{x}_j^z + (\tau + 1)\bar{y}_{ij}^z \bar{A}_{ki}^w &\leq \tau & \forall j \in \mathcal{V}^w, \\ & & \forall k \in \mathcal{N}(i) \setminus \{j\} \end{aligned} \right\} \quad (29)$$

$$\left. \begin{aligned} \sum_{j \in \mathcal{V}^w} A_{vj}^w &\leq M^z c_v \\ \sum_{j \in \mathcal{V}^w} A_{vj}^w &\geq \varepsilon^z c_v \\ s_v^{\text{clean}} &\leq c_v \\ s_v^{\text{clean}} &\leq s_v \\ s_v^{\text{clean}} &\geq s_v + c_v - 1 \end{aligned} \right\} \quad \forall v \in \mathcal{V}^w \quad (30)$$

Computational efficiency can be improved using symmetry constraints:

$$y_{ii}^z + \bar{y}_{ii}^z \leq 1 \quad \forall i \in \mathcal{V}^w \quad (31)$$

$$s_i - (1 - A_{ii}^w) \geq 0 \quad \forall i \in \{0, \dots, |\mathcal{E}|\} \quad (32)$$

4. EXAMPLE

The optimization program presented in Sec. 3 has been implemented with the Python API to Gurobi, `gurobipy` in Python 12 for the example network Apulia (24 nodes, 34 links, 1 reservoir) (Hall, 2021) and solved using the Gurobi 12.0 branch-and-bound solver for MINLP. The network has been adjusted to contain no parallel edges prior to the analysis. Pumps and valves have been substituted by simple links. The preprocessing was conducted with the Python package WNTR (Klise et al., 2017).

The result is shown in Fig. 3. Three out of six boundary edges have been divided and 8 sensors have been placed

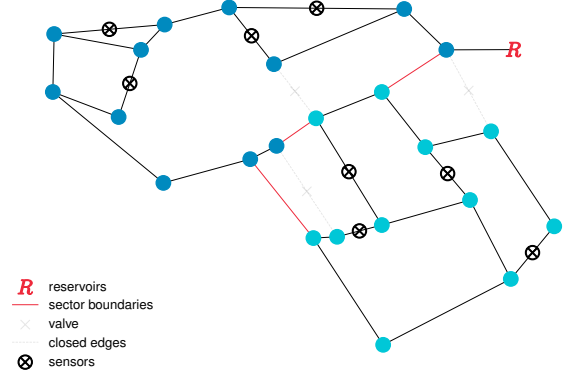


Fig. 3. Optimization result for the example network Apulia, divided into 2 sectors indicated by different node colors

in the network. Notably, each cycle of the network graph contains at least 1 sensor.

The program took approximately 16s to solve within a Windows Subsystem for Linux (WSL2) environment using Ubuntu 24.04 LTS on a Windows 10 Pro machine equipped with an AMD EPYC 7313 16-Core Processor @ 3.00GHz and 64GB of RAM. Constructing the constraints took significantly longer (about 160s) than solving. The model has 383564 quadratic constraints, 34 simple general constraints, 34 function constraints treated as nonlinear, 218 continuous variables, and 14698 integer variables out of which 14582 are binary.

5. DISCUSSION AND CONCLUSION

This work has demonstrated how the structural observability theory for WDSs developed by van Gemert et al. (2024) can be used in order to include a novel monitoring metric in the dividing stage of sectorization. The dividing stage has been modelled as a MINLP to which the computation of the monitoring metric, i.e. sensor placement problem, has been integrated.

The proposed methodology requires further validation on larger benchmarks such as the L-TOWN network to assess the practicability of the results. Notably, it has been shown that structural observability methods underestimate the required number of required sensor nodes (Montanari and Aguirre, 2020). However, we believe that the metric can still be useful as a proxy for how well the system can be monitored due to the simple and intuitive formulation. It can also be easily adjusted to prefer some sensor locations to others by introducing weights to the problem formulation.

Another limitation of the proposed methodology is scalability due to the large problem size, especially the number of constraints. This issue could be improved to some degree with further modelling adjustments. Lastly, the MINLP can be extended to consider multiple timesteps as has been demonstrated recently on the problem of pump scheduling by Henn and Pelz (2025).

Parameter	Domain	Description
$Q_{\max}$	$\mathbb{R}^+$	maximum volume flow in pipes
$Q^{t,\max}$	$\mathbb{R}^+$	maximum volume flow in tank
Q_v	$\mathbb{R}^+$	demand at node v
α_v	$\mathbb{R}^+$	elevation at node v
$\Delta P^{\max}$	$\mathbb{R}^+$	maximum head loss
f_{ij}, d_{ij}, l_{ij}	$\mathbb{R}^+$	friction coeff., diameter, length of edge ij
S_t	$\mathbb{R}^+$	surface area of tank t
$P_t^0, P_t^{\max}$	$\mathbb{R}^+$	initial and maximum pressure (level) in tank t
β_{ij}	$\{0, 1\}$	indicator of boundary edges
α^{inc}	$\{0, 1\}$	incidence matrix of WDS graph
τ	$\mathbb{Z}^+$	propagation time for zero forcing
M^z, ε^z	$\mathbb{R}^+$	big-M parameters for zero forcing
Variable	Domain	Description
b_{ij}	$\{0, 1\}$	indicator of edge division
q_{ij}	$[-Q_{\max}, Q_{\max}]$	volume flow
Δp_{ij}	$[0, \Delta P_{\max}]$	pressure loss
p_i	$[0, \infty]$	pressure
x_{ij}, y_{ij}	$\{0, 1\}$	flow direction indicators
ϱ_t	$\{0, 1\}$	tank inflow/outflow indicator
B^o	$\{0, 1\}$	open boundary edge matrix
A^{inc}	$\{0, 1\}$	incidence matrix of the associated graph
$A^w, \bar{A}^w$	$\{0, 1\}$	system matrix, c.f. Eq. 7
s_v	$\{0, 1\}$	sensor placement indicator
s_v^{clean}	$\{0, 1\}$	sensor placement indicator without sensors on closed edges
$x_v^z, \bar{x}_v^z$	$\{0, \dots, \tau\}$	time at which node v is forcing
$y_{ij}^z, \bar{y}_{ij}^z$	$\{0, 1\}$	indicator whether edge ij forced
c_v	$\{0, 1\}$	indicator whether a node in the associated graph connects to other nodes
$I, \bar{I}$	$\{0, 1\}$	identity matrices, c.f. Eqs. 7 and 4
$D, \bar{D}$	$\{0, 1\}$	discharge matrices, c.f. Eqs. 7 and 4
Set	Description	
$\mathcal{E}, \mathcal{V}$	WDS graph edges and nodes	
$\mathcal{J}$	WDS junctions	
$\mathcal{T}$	WDS tanks	
$\mathcal{V}^w$	associated graph nodes	
$\mathcal{N}(i)$	out-neighbors of node i	

Table 1. Specification of parameters, variables and sets in the MINLP in Sec. 3

ROLES AND CONTRIBUTIONS

Michaela Leštáková: Conceptualization, Data curation, Formal analysis, Methodology, Investigation, Software, Visualization, Writing – original draft, Writing – review & editing; **Peter F. Pelz:** Supervision, Project administration, Funding acquisition.

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